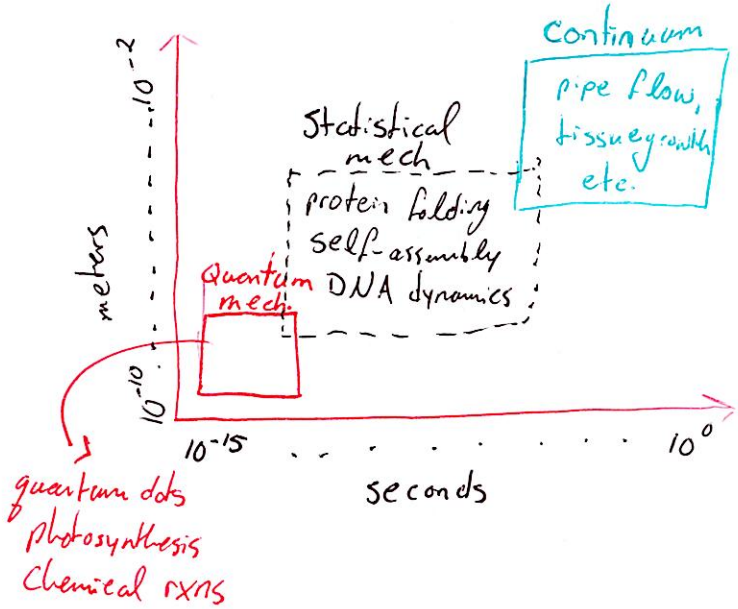
The "Zen" of Computational modeling + materials simulation.



*Adapted from Prof. Gaurav Arya Lecture Slides

We want to relate <u>microscopic properties</u> to <u>macroscopic observables</u>.

For observable A: we measure the ensemble average $\langle A \rangle$ which is the weighted average: $\langle A \rangle = \sum_{\nu} A_{\nu} P_{\nu}$

Postulate: The ensemble average and long-time average are identical

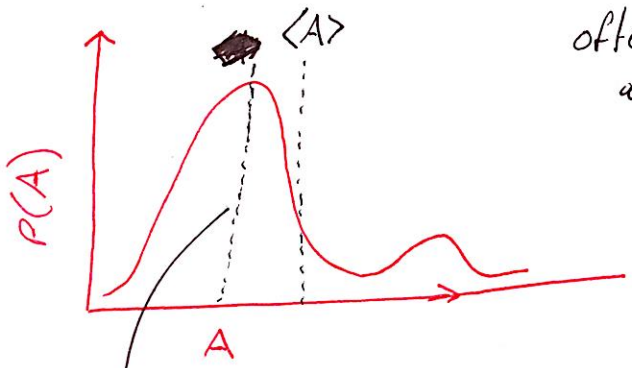
$$\langle A \rangle = \lim_{t \to \infty} \frac{1}{t} \sum_{i} A_i \Delta t_i$$

Therefore we can either watch multiple ~~states~~ systems at once, or one system for a long time.

In the end, we want the <u>ensemble</u> of states that defines a macroscopic observable.

↳ <u>we want to approximate the partition function</u>

What are the implications of this?



experimental structure
i.e. crystal structure
or cryo-EM

often our experimental observable in bulk assays will not be representative of underlying properties of our system.

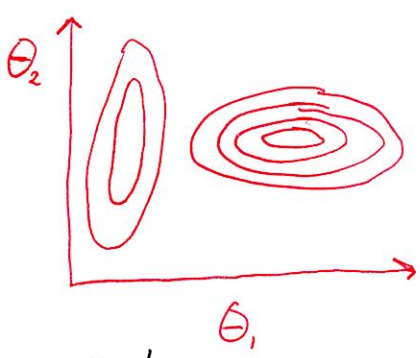
How to get around this?

↳ Single molecule experiments

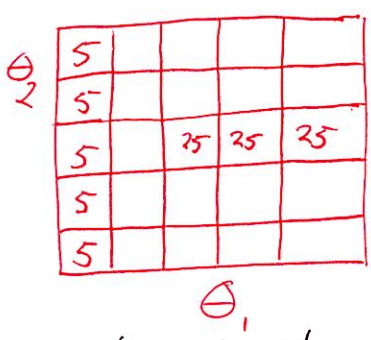
↳ Computational prediction of "ensemble"

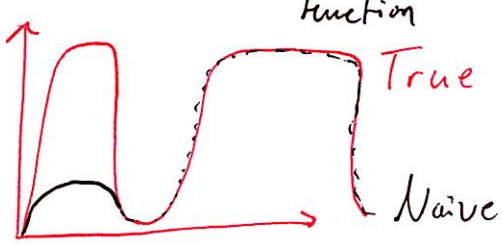
How can we predict the underlying probability distribution?
How can we analyze our results?
How can we interpret our results?

Thought exercise: Adapted From Prof. Daniel Zuckerman "Maximum Likelihood vs. Bayesian estimation of uncertainty"



Continuous



discretized likelihood (%)

Q: What does a probability density function projected along θ_1 look like?